2019 HCI H2 PHYSICS Prelim P3 Suggested Solutions

1	(a)	(i)	The gas molecules rebound off the incoming piston at speed higher than before the collisions. The increase in average KE corresponds to an increase in temperature.	B1
		(ii)	The frequency of collisions increases. The change in momentum per collision is higher.	B1 B1
	(b)	(i)	For ideal gas, total internal energy = total number of molecules x mean kinetic energy of a molecule of ideal gas $U = \frac{3}{2}NkT$ $\Delta U = \frac{3}{2}Nk\Delta T$ $\Delta U = \frac{3}{2}\Delta(pV)$ $\Delta U = \frac{3}{2}(30 \times 20 - 10 \times 40)(10^5 \times 10^{-6})$ $= 30$ J	M1 A1
		(ii)	During the change B $\rightarrow$ C $\rightarrow$ A, By First Law of Thermodynamics, $W = 0 + (-p\Delta V)$ $= -(10)(40 - 20)(10^5 \times 10^{-6})$ $= -20$ J $\Delta U = Q + W_{on}$ $-30 = Q + (-20)$ $Q = -10$ J	M1 M1 A1